

6.2 Final Project Reference Manual: Chemometric Data Reduction & Diagnostic Spectral Profiling

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1 Introduction & Laboratory Context

In environmental pollution tracking and microplastic screening, raw infrared spectra collected from field samples rarely align cleanly with pristine laboratory reference databases. Environmental exposure introduces several overlapping confounding factors: organic biofouling layers (biofilms), surface photo-oxidation, and baseline scattering distortions caused by irregular fragment boundaries.

This handbook accompanies **Module 6-2**, which provides an automated workflow to decouple these multi-component variations without operator bias. Using authentic micro-FTIR data from nylon fragments exposed to the **Mekong River in Phnom Penh, Cambodia**, we analyze how sample preservation pathways (Wet vs. Dry storage) change the efficiency of advanced chemical oxidation techniques (H₂O₂ wet digestion).

2 The Mathematical Preprocessing Pipeline

2.1 1. Continuous Linear Interpolation (Spectral Alignment)

Fourier-Transform Infrared (μ FTIR) instruments frequently experience mechanical wavelength adjustments or minor source window shifts between calibration runs, causing variations in the absolute wavenumber index steps. Comparing rows across unaligned matrices introduces index-shifting artifacts.

To establish a uniform, column-matched feature space, the pipeline identifies the absolute overlapping coordinate boundaries ($x_{\min}, x_{\max}$) and maps all raw spectrum arrays onto a standard vector of exactly 1,000 variables using continuous linear interpolation:

$$Y_{\text{interp}}(x) = Y_1 + \left(\frac{x - x_1}{x_2 - x_1} \right) \cdot (Y_2 - Y_1) \quad (1)$$

This operation guarantees that subsequent matrix calculations compare identical physical chemical bands across all samples.

2.2 2. Z-Score Mean Matrix Standardization

Variations in microplastic particle thickness, path lengths, or localized beam focus cause global vertical offsets in absorbance intensity. To prevent absolute baseline intensity from biasing the variance analysis, data is normalized using a standard Z-score transformation:

$$Z_{ik} = \frac{X_{ik} - \mu_k}{\sigma_k} \quad (2)$$

where μ_k represents the mean intensity of wavenumber column k and σ_k is its standard deviation. This shifts the model’s focus from global scaling variations to localized peak changes.

3 Unsupervised Feature Extraction Frameworks

3.1 1. Global Linear Deconvolution via PCA

Principal Component Analysis (PCA) reduces the dimensionality of the 1,000-variable spectral dataset by finding orthogonal axes that maximize global variance. Principal Component 1 (PC1) typically captures broad, macro-scale variations like baseline offsets. Consequently, **Principal Component 2 (PC2)** holds the key structural variances that separate the experimental treatments.

3.2 2. Automated Peak Extraction via Loading Signatures

Instead of relying on manual peak selection, the code uses the first derivative of the PC2 loading vector to locate local maxima and minima automatically via the peak condition rule:

$$\left| \frac{\partial \text{PC2}_{\text{Loading}}}{\partial \bar{\nu}} \right| = 0 \quad \text{subject to } |\text{Weight}| \geq 0.05 \quad (3)$$

The algorithm automatically flags two main chemical indicators in the Mekong River dataset:

- **The 1663.5 cm^{-1} Coordinate Vector:** Corresponds to the Amide I structural protein band from biological biofouling.
- **The 3317.5 cm^{-1} Coordinate Vector:** Corresponds to the Hydroxyl ($-\text{OH}$) stretching envelope.

This provides definitive chemical proof that pre-drying environmental samples at 40°C alters the surface boundary layer, trapping biofilm residues and reducing the effectiveness of peroxide oxidation compared to wet storage.

3.3 3. Non-Linear Neighborhood Mapping (t-SNE)

While PCA rotates coordinate axes to preserve large-scale global distances, **t-SNE (t-Distributed Stochastic Neighbor Embedding)** acts as a non-linear projector that prioritizes local neighborhoods.

It converts Euclidean distances between spectra into localized conditional probabilities, mapping similarities into tight, isolated clusters. This approach reveals regional patterns and treatment subgroups that linear projections can miss.

4 Quick-Reference Pipeline Execution Syntax

To evaluate a new set of field samples, ensure your raw spectrum documents are inside your workspace path and adjust your script configuration parameter block:

```
1 # Final Project Code Block
2 folders = {
3     'Pristine': 'Nylon_irregular_pristine/',
4     'Biofouled_Wet': 'Nylon_irregular_biofouled_Mekong_wet/',
5     'Biofouled_Dry': 'Nylon_irregular_biofouled_Mekong_dry/'
6 }
7
8 # The pipeline automatically ignores background calibrations ('bkg')
```

9 *# and standardizes the output feature array to a uniform grid*

Listing 1: Adapting Data Loaders for Final Project Evaluation